Total Printed Pages: 02

Roll No. .....

MAB-101

Examination –june- 2026

B.Tech. I Sem : All braches

Linear algebra and calculus

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any. Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

.1 (a) Write the statement of Taylor's theorem Theorem — 02

(b) Write the cartesian formula for radius of curvature — 02

(c) Find the radius of curvature of the following curves at the points indicated against them $\sqrt{x}+\sqrt{y}=\sqrt{a}, \left(\frac{1}{4},\frac{1}{4}\right)$ — 03

(d) Discuss the maxima and minima of the function $u = \sin x \sin y \sin(x+y)$ — 05

OR

Find the radius of curuature at the Point "t" of the curve. $x = 3a\cos t - a\cos 3t$, $y = 3a\sin t - a\sin 3t$ — 05

(a) Write the definition of Homogeneoue function — 02

(b) State and prove Euler's theorem — 02

(c) $U = \log\frac{(x^4+y^4)}{(x+y)}$ then find the value of $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$ — 03

(d) (d) If u = f(r), Where $r^2 = x^2 + y^2$ Show that $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 = f''(r) + 1/r\, f'(r)$. — 05

OR

If $u = \tan^{-1}\frac{x^3+y^3}{x-y}$ then show that — 05

$$x^2\frac{\partial^2 u}{\partial x^2}+y^2\frac{\partial^2 u}{\partial y^2}+2xy\frac{\partial^2 u}{\partial x\partial y}=2\cos 3u.\sin u$$

(a) Write the fomula of definite integral as limit of sum — 02

(b) Find the limit as n→∞ of the series $\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}+\cdots+\frac{1}{2n}$ — 02

(c) Find the limit as n→∞ of the series — 03

$$\lim\left[(1+1/n)(1+2/n)(1+3/n).....(1+n/n)\right]^{\frac{1}{n}}$$

(a) curvature is given by

05

(d) *Evaluate*

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dzdydx}{(1+x+y+z)^2}$$

OR

05

Evaluate $\iint_R e^{2x+3y}\, dx\, dy$ over the triangle bounded by $x=0$, $y=0$, $x+y=1$

02

Q.4 (a) Find the characteristic root's of the matrix $A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$

02

(b) Define Normal form of a matrix with example

03

(c) Reduce the Matrix to normal form and find it`s rank.

$$A=\begin{bmatrix}1 & -1 & 2 & -1\\ 4 & 2 & -1 & 2\\ 2 & 2 & -1 & 0\end{bmatrix}$$

05

(d) Check the consistency of the following system of equations.

$$7x_1+2x_2+3x_3 = 16$$
$$2x_1+11x_2+5x_3 = 25$$
$$x_1+3x_2+4x_3 = 13$$

OR

05

Determine the eigen values and eigen vectors of the matrix

$$A=\begin{bmatrix}6 & -2 & 2\\ -2 & 3 & -1\\ 2 & -1 & 3\end{bmatrix}$$

Q.5 (a) Define logically equivelence

02

(b) If p and q are two statements prove that $(p \wedge q) \Rightarrow q$ is a tautology

02

(c) State and prove Idempotent law

03

OR

(d) Express the following function into conjuctive normal form:
F(x, y, z) =(x+y+z)(xy+x'y+x'z)'

05

OR

Define (1) Graph (2) Degree of vertex (3) path (4) tree (5) loop

05